puts things in formation. Horoscopes put lovers in astrological formations, propaganda broadcasts put voters in political formations and marching songs put soldiers in military formations.

As a paradigmatic case, consider music. Most symphonies, melodies and tunes don't represent anything, which is why it makes no sense to ask whether they are true or false. Over the years people have created a lot of bad music, but not fake music. Without representing anything, music nevertheless does a remarkable job in connecting large numbers of people and synchronising their emotions and movements. Music can make soldiers march in formation, clubbers sway together, church congregations clap in rhythm and sports fans chant in unison.[11]

The role of information in connecting things is of course not unique to human history. A case can be made that this is the chief role of information in biology, too.[12] Consider DNA, the molecular information that makes organic life possible. Like music, DNA doesn't represent reality. Though generations of zebras have been fleeing lions, you cannot find in the zebra DNA a string of nucleobases representing 'lion' nor another string representing 'flight'. Similarly, zebra DNA contains no representation of the sun, wind, rain or any other external phenomena that zebras encounter during their lives. Nor does DNA represent internal phenomena like body organs or emotions. There is no combination of nucleobases that represents a heart or fear.

Instead of trying to represent pre-existing things, DNA helps to produce entirely new things. For instance, various strings of DNA nucleobases initiate cellular chemical processes that result in the production of adrenaline. Adrenaline too doesn't represent reality in any way. Rather, adrenaline circulates through the body, initiating additional chemical processes that increase the heart rate and direct more blood to the muscles.[13] DNA and adrenaline thereby help to connect trillions of cells in the heart, legs and other body parts to form a functioning network that can do remarkable things, like run away from a lion.

If DNA represented reality, we could have asked questions like 'Does zebra DNA represent reality more accurately than lion DNA?' or 'Is the DNA of one zebra telling the truth, while another zebra is misled by her fake DNA?' These, of course, are nonsensical questions. We might evaluate DNA by the fitness of the organism it produces, but not by truthfulness. While it is common to talk about DNA 'errors', this refers only to mutations